

What is Bayesian Statistics?

→ Bayesian statistics updates our understanding of some quantity through combining based on prior knowledge and new data

Bayes'

Theorem

$$\text{Posterior} = \frac{(\text{likelihood})(\text{Prior})}{\text{Evidence}}$$

$$\text{or } p(\theta | y) = \frac{p(y | \theta) \cdot p(\theta)}{p(y)}$$

θ = Model parameters, like a scaling factor ex. $s(\theta)$ value

y = observed data like experimental s factor

$p(\theta, y)$ = Posterior, what we now believe about θ after seeing the data

↳ What's the prob of parameter θ given the observed data y ?

$p(y, \theta)$ Likelihood, how well the model predicts data for a given θ

↳ What is the prob of observing data y , given the model parameters?

$p(\theta)$ = Prior, Our belief about θ before seeing the data

$p(y)$ = Evidence, a normalization factor

Why use Bayesian Methods in nuclear astro?

Traditional methods

- tend to crudely combine uncertainties, causing info to be lost in translation
- struggle with dealing with outliers and inconsistent datasets
- assumes Gaussian, symmetric errors which become especially problematic when values like cross sections can't be negative

Bayesian models allow for:

- non-Gaussian errors
- producing full prob distributions for quantities like S-factors, instead of just best fit values.

Paper 1: Iliadis et al. (2016) - Bayesian Estimation of Thermonuclear Reaction Rates

paper introduces Bayesian framework to improve how we estimate astrophysical s factors & ThermoNuc RR

↳ addresses key limitations of traditional methods

4 core methods

① Bayesian modeling of s factors
 s factor hella smooth when it comes to energy dependence as we know at low energies Bayesian models are used to fit $S(E)$ using polynomials (or nuclear theory??) but now also estimating a full prob distribution for it instead of just a best fit

MODEL INCLUDES:

→ Experimental s factor values with uncertainties

→ s factor vs energy graph

② incorporating uncertainties

→ Statistical uncertainties are modeled with Gaussian

→ Systematic uncertainties modeled with 'lognormal priors'

↳ smthg like detector efficiency, whole system is impacted, tries to model this overall uncertainty

Outlier treatment: Robust Regression

- ③ Each data point is modeled as possibly coming from 1 of 2 populations
- (a) Good data with trustworthy uncertainties
 - (b) Outliers with underestimated uncertainties

Bayesian stats automatically reduce the weight of outlier points without throwing them out

- ④ Non-Gaussian uncertainty modeling
- Low-yield non symmetric at times, avoids unphysical negative values

- ⑤ MCMC with JAGS
- JAGS is a program that performs Bayesian analysis by generating random samples from the posterior distribution of model parameters, using MCMC method.

MCMC $\rightarrow$ way to explore complex prob distributions by taking many small random steps (Random walk) to figure out what parameter values best fit the data set.

Paper used $d(p, \gamma)$ the reaction, key result is Bayesian analysis either matched or improved on results

$\rightarrow$ provided more realistic error bands